

Risk Across Tails and Timescales

RESEARCH QUESTION

Does tail dependence change with the investment horizon, and what does ignoring it do to a portfolio's measured risk?

DATASET

Daily adjusted closing prices of six US sector ETFs (XLE · XLF · XLK · XLV · XLI · XLU), Yahoo Finance, 1999–2026; equal-weight portfolio; in-sample to 2019, out-of-sample 2020–2026

METHOD

GARCH-t margins, MODWT wavelet horizon bands and copulas; 1-day and 10-day VaR / ES backtests

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Executive summary

Question. Does tail dependence between US equity sectors change with the investment horizon, and what does ignoring that do to a portfolio's measured risk?

Answer (a): the crash tail does not fade, the rally tail does. Using a MODWT wavelet decomposition of GARCH-filtered returns (1999–2019), the probability that two sectors crash together (χ_L , $q = 5\%$) is 0.36 at the 2–8 day scale and 0.31 beyond 32 days, a change of -0.05 that is not significant ($p = 0.212$). Rally co-movement falls from 0.35 to 0.20 ($p < 0.001$). So dependence becomes *asymmetric* at long horizons, and at every scale crash co-movement exceeds what a Gaussian copula implies (by +0.06 to +0.10). A symmetric t-copula reads the fading rally tail as fading tail dependence (λ 0.14 $\rightarrow$ 0.05) and would understate long-horizon crash risk.

Answer (b): ignoring it understates risk, most at multi-day horizons. Out-of-sample (2020–2026, 1,699 days), the Gaussian-copula benchmark gives 21 breaches of 99% VaR against 17 expected (t-copula: 17) and a daily ES 97.5% 4.2% lower than the t-copula. At the 10-day horizon the Gaussian path model's ES 97.5% is 12.2% below the horizon-aware wavelet empirical-copula model (14.9% in stressed windows); aggregating a daily t-copula still leaves 10.1%. Getting volatility right matters most: every GARCH model beats historical simulation (Diebold–Mariano $p \leq 0.007$).

Recommendation. Replace the Gaussian copula with a t-copula on daily-updated GARCH margins for the daily limit, and set the 10-day ES from the horizon-matched wavelet model instead of $\sqrt{10} \times 1$ -day: today that is 6.01% of NAV. Over 2020–2026 the $\sqrt{10}$ shortcut sat below this model in 76% of windows. Details in Section 6.

1. Design choices and why

Table 1 sets out each design choice, the reason for it and the alternative considered. Each component answers one part of the question and is tested out of sample against a simpler alternative.

Choice	Specification	Rationale	Alternative considered
Portfolio	Equal weight, 6 Select Sector SPDRs	Sectors are where diversification is supposed to come from; equal weight keeps the result about dependence, not weights	Adding SPY: it contains the sectors and double-counts beta
Market & data	US, Yahoo Finance adjusted closes, 1999-01-04 to 2026-10-06 (6982 days)	Deep, liquid, synchronous trading; covers dot-com, GFC, COVID and 2022	Sri Lankan CSE: thin trading biases tail estimates
Split	Estimate $\leq$ 2019, test 2020 $\rightarrow$ today	The test window contains a crash (COVID) and a rate shock (2022)	Random split: leaks future information
Margins	GARCH(1,1)-t, volatility updated daily; FHS shocks	Removes volatility clustering so the copula sees dependence, not shared volatility (McNeil & Frey 2000)	Raw returns: inflate measured tail dependence
Horizon tool	MODWT multiresolution analysis, LA(8), $J = 6$, bands H1 2–8d, H2 8–32d, H3 $>32d$	Shift-invariant, any sample length, additive (bands sum to the series)	DWT (needs 2^J length); overlapping h-day returns (autocorrelated)
Dependence	Gaussian, Student-t, Clayton (full likelihood, AIC); empirical copula; χ_L , χ_U	Gaussian = no tail dependence; t = symmetric; Clayton = crash-only; empirical = no shape imposed	Composite pairwise likelihoods: not comparable across families
Risk measures	VaR 95/99 %, ES 97.5 % (FRTB), at 1 and 10 days	VaR for Basel backtesting; ES because tail dependence matters beyond the quantile	VaR only: blind to tail shape
Benchmarks	Gaussian copula (same margins); historical simulation	Isolates the dependence choice; HS is the industry default	No benchmark

Table 1. Design choices. Every choice is a constant in `src/config.py`.

2. Data, exploration and preprocessing

Data quality. The bundled file holds 6,982 trading days from 1999-01-04 to 2026-10-06 for all six ETFs, with 0 missing prices, 0 duplicate dates and 0 non-positive prices, so no imputation or filtering is needed. Prices are dividend-adjusted closes; we work with daily log returns for modelling and simple returns for portfolio profit and loss. The largest daily moves (for example 22.5% for XLE on 2020-03-09) are genuine market events and are kept: they are exactly the observations a tail-risk study is about.

Series	Ann. return	Ann. vol.	Skew	Excess kurtosis	Worst day	Jarque–Bera p	Ljung–Box r^2 p	ARCH-LM p
XLE	6.6%	26.8%	-0.40	8.9	-15.6%	<0.001	<0.001	<0.001
XLF	4.3%	29.3%	-0.05	14.5	-18.2%	<0.001	<0.001	<0.001
XLK	6.0%	25.3%	0.20	6.2	-9.1%	<0.001	<0.001	<0.001
XLV	7.9%	17.8%	-0.16	8.1	-10.3%	<0.001	<0.001	<0.001
XLI	7.6%	20.7%	-0.27	5.6	-9.9%	<0.001	<0.001	<0.001

Series	Ann. return	Ann. vol.	Skew	Excess kurtosis	Worst day	Jarque–Bera p	Ljung–Box r^2 p	ARCH-LM p
XLU	7.3%	18.3%	0.02	8.7	-8.9%	<0.001	<0.001	<0.001
Portfolio	7.6%	18.6%	-0.21	9.6	-10.4%	<0.001	<0.001	<0.001

Table 2. Daily log returns, 1999–2019 (estimation sample). Jarque–Bera tests normality; Ljung–Box on squared returns (10 lags) and ARCH-LM (5 lags) test for volatility clustering.

What the data show. Every sector has excess kurtosis between 5.6 and 14.5 and every normality test rejects, so a normal margin would understate tail losses. Squared returns are strongly autocorrelated (all $p < 0.001$): calm and turbulent periods cluster. Dependence is not constant either: the trailing one-year average correlation between sectors ranges from 0.13 to 0.85 and moves with portfolio volatility (correlation 0.54), so diversification is weakest exactly when it is needed (Figure 1). On days when one sector has a worst-5% day, on average 2.1 of the other five do too, against 0.25 if they were independent.

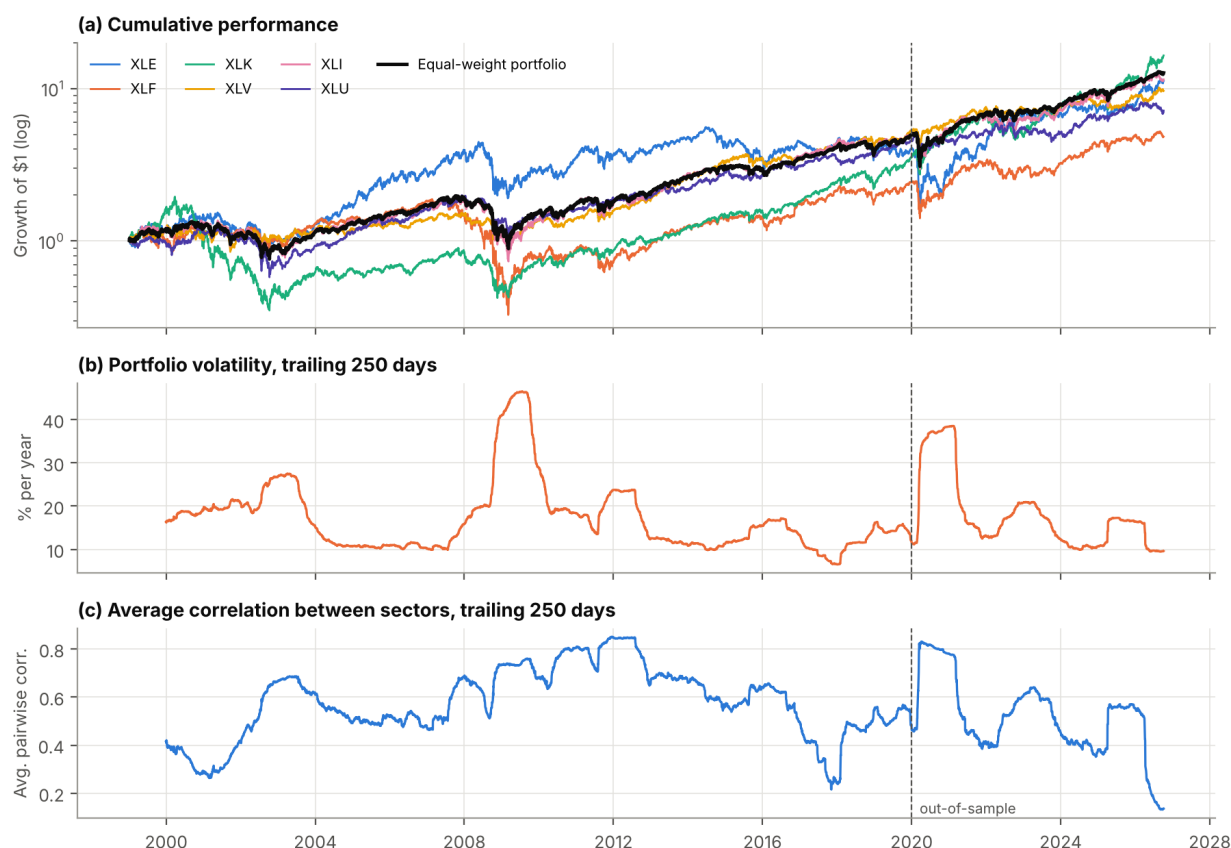


Figure 1. (a) Growth of \$1 in each sector and the equal-weight portfolio (log scale); (b) portfolio volatility; (c) average pairwise correlation. Dashed line: start of the out-of-sample period.

Preprocessing decisions. (1) Model each sector with GARCH(1,1) and Student-t innovations, because returns are fat-tailed and volatility clusters. (2) Measure dependence on the GARCH-standardised residuals, not on raw returns, so that shared volatility is not mistaken for tail dependence. Figure 2 confirms the filter works: the autocorrelation of squared residuals falls inside the no-autocorrelation band, residual excess kurtosis drops to 0.7–2.2, and the Ljung–Box test on squared residuals no longer rejects for 5 of 6 sectors (utilities keep mild clustering, $p = 0.000$). GARCH persistence $\alpha + \beta$ is 0.987–0.999. (3) Convert residuals to ranks for the copulas, so the dependence estimate does not depend on the margins. (4) Re-estimate every 20 days on a rolling window, because Figure 1 shows dependence drifting over time.

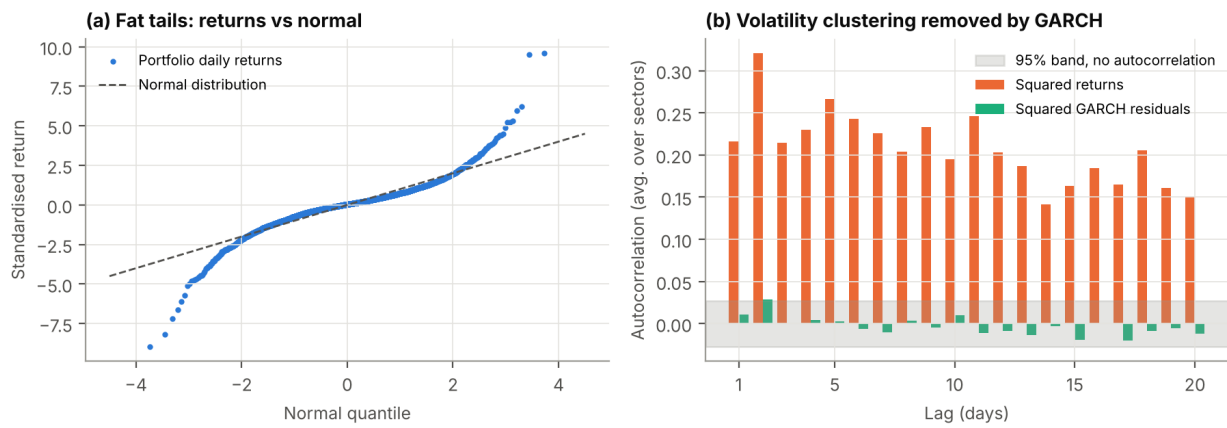


Figure 2. (a) Standardised portfolio returns against the normal distribution, 1999–2019; (b) autocorrelation of squared returns before and after GARCH filtering, averaged over the six sectors.

3. Method

Margins. For each sector, $r_t = \mu + \sigma_t z_t$ with $\sigma_t^2 = \omega + \alpha(r_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2$ and Student-t z . Out-of-sample the parameters are re-estimated every 20 trading days on the trailing 1000 days, and σ_t is rolled forward every day with the realised return, so a forecast for day t uses information up to $t - 1$ only. Simulated shocks are drawn from the empirical distribution of the standardized residuals (filtered historical simulation).

Wavelets. The MODWT multiresolution analysis splits each residual series into details D_j (fluctuations on scales of 2^{j-1} – 2^j days) and a smooth S_6 , with $\sum_j D_j + S_6$ equal to the series exactly (computed in the frequency domain; the test suite checks additivity). We group them into H1 = D1+D2, H2 = D3+D4 and H3 = D5+D6+S6, reflect the series at the ends and drop boundary-affected values in the in-sample analysis.

Tail co-movement. For each pair of sectors, $\chi_L(q) = P(U_j \leq q \mid U_i \leq q)$ is the probability that one sector is in its worst q -tail given the other is (χ_U : best tail). As $q \rightarrow 0$ it becomes the tail-dependence coefficient λ . We average over the 15 pairs and compare with the value a Gaussian copula with the same correlation implies. Confidence intervals and tests use a stationary block bootstrap (400 replications, mean block 60 days) that re-runs the wavelet decomposition on every resample. Copulas are fitted to rank pseudo-observations by maximum likelihood with the full d -dimensional density, so AIC compares like with like.

Risk. 1-day VaR and ES come from 20,000 Monte Carlo draws of the copula, mapped to shocks and scaled by each sector's σ_t . 10-day ES uses 10,000 simulated GARCH paths (buy-and-hold). To isolate what each simplification costs, we change one thing at a time: the $\sqrt{10}$ rule, the copula family, how the 10-day dependence is formed (aggregated daily paths vs a copula imposed at the horizon), and which scale the copula is estimated on (daily vs the H2 band that contains 10 days).

Validation. VaR: Kupiec coverage, Christoffersen independence, Engle–Manganelli dynamic quantile, Basel traffic light. ES: Acerbi–Szekely Z2 and the McNeil–Frey exceedance test. Model comparison: the FZ0 joint VaR–ES loss (Patton, Ziegel & Chen 2019) with Diebold–Mariano tests. Monte Carlo error is checked by re-running the whole out-of-sample engine with 3 more seeds.

4. Does tail dependence change with the horizon?

Series	Scale	AIC winner	t-copula ν	t-copula λ	Δ AIC Gaussian	Δ AIC Clayton
Daily residuals	1 day	student	8.1	0.15	1,079	6,555
H1	2–8 days	student	9.6	0.14	640	6,333
H2	8–32 days	student	8.9	0.15	766	6,178
H3	> 32 days	student	16.1	0.05	209	4,833

Table 3. Full-likelihood copula fits, 1999–2019. Δ AIC is relative to the winner (larger = worse). The t-copula wins at every scale; its symmetric λ falls at the longest scale.

The t-copula is preferred at every scale by a wide AIC margin, so tails matter. Its tail-dependence coefficient is stable from daily data to H2 (0.14–0.15) and drops at H3 (0.05, $\nu = 16$). Taken at face value, tail dependence fades at long horizons. But a t-copula forces the crash and rally tails to be equal. Figure 3 measures them separately.

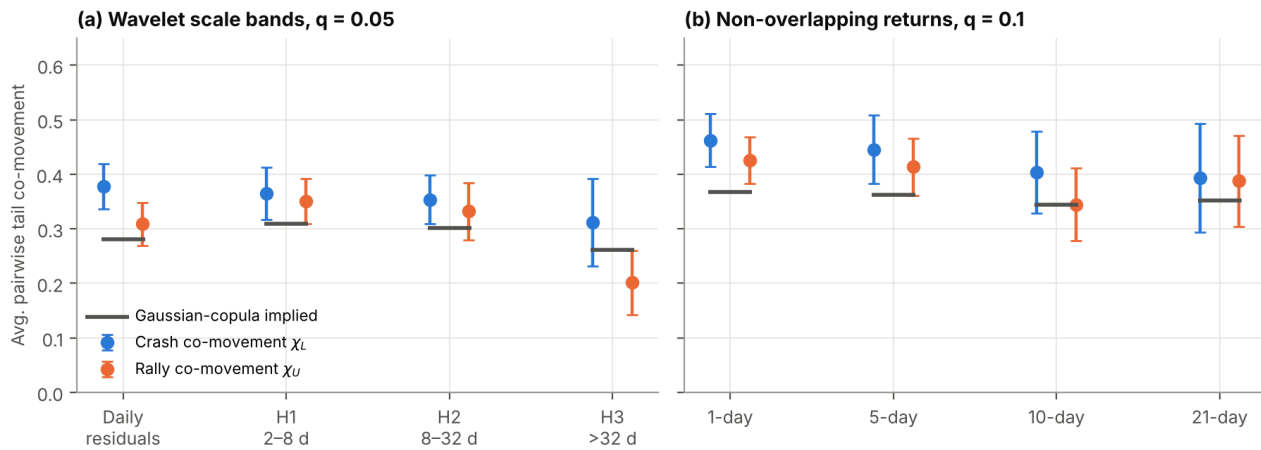


Figure 3. Average pairwise crash (χ_L) and rally (χ_U) co-movement with 95% bootstrap intervals; bars mark the Gaussian-copula value at the same correlation. (a) wavelet bands of GARCH-filtered residuals; (b) non-overlapping raw h-day returns, 1999–2019.

Test (q = 5%, wavelet bands)	Estimate [95% CI]	p-value	Verdict
Crash co-movement, H3 – H1	-0.053 [-0.136, +0.030]	0.212	no evidence of change
Rally co-movement, H3 – H1	-0.149 [-0.216, -0.081]	<0.001	different
Asymmetry $\chi_L - \chi_U$, H1	+0.014 [-0.030, +0.058]	0.526	symmetric
Asymmetry $\chi_L - \chi_U$, H3	+0.110 [+0.012, +0.209]	0.029	different
Crash co-movement above Gaussian, H1	+0.056 [+0.025, +0.087]	<0.001	above Gaussian
Crash co-movement above Gaussian, H3	+0.051 [-0.020, +0.122]	0.157	not distinguishable

Table 4. Bootstrap tests (400 stationary-bootstrap replications re-running the MODWT).

Finding. Crash co-movement does not decline significantly with scale (H3 – H1 = -0.053, $p = 0.212$), while rally co-movement roughly halves ($p < 0.001$). At short scales the two tails are symmetric; beyond a month crash co-movement exceeds rally co-movement by 0.11 ($p = 0.029$). This is the long-horizon version of the asymmetric correlation found by Longin & Solnik (2001) and Ang & Chen (2002): sectors rally separately but fall together, and that remains true when the holding period lengthens. In raw h-day returns (panel b) the picture is noisier (only 251 independent 21-day observations exist), but crash co-movement stays above the Gaussian level up to 10 days.

Horizon	Observed χ_L [95% CI]	Gaussian model	Student-t model	Empirical (FHS) model
1 day	0.462 [0.414, 0.511]	0.379	0.402	0.436
5 days	0.446 [0.383, 0.508]	0.381	0.392	0.407
10 days	0.404 [0.329, 0.479]	0.377	0.384	0.398
21 days	0.393 [0.293, 0.493]	0.371	0.381	0.389

Table 5. Crash co-movement of h-day returns ($q = 10\%$): observed vs implied by daily GARCH + copula models simulated forward (60,000 paths each), 1999–2019.

Table 5 asks whether a model estimated on daily data reproduces this. At one day the Gaussian model implies 0.379 and the t-copula 0.402, both below the 95% interval of the observed 0.462; only the empirical copula (0.436) lies inside it. At longer horizons every model stays below the observed value, though the intervals widen and overlap. **Ignoring tail dependence therefore means understating joint crashes, and the shortfall does not disappear as the horizon lengthens.** Section 5 measures what that does to VaR and ES.

5. What does ignoring it do to measured risk?

5.1 One-day risk, out-of-sample 2020–2026

Model	99% VaR breaches	Kupiec p	Indep. p	DQ p	Basel red / yellow	Mean ES 97.5%	Z2 p	Breach loss / ES	FZ0 loss
Historical simulation (250d)	25 (25–25)	0.068	<0.001	<0.001	4% / 24%	3.23%	0.011	1.13	-3.323
Filtered HS (empirical copula)	12 (12–14)	0.199	0.071	0.010	0% / 2%	2.96%	0.993	1.08	-3.625
Gaussian copula (benchmark)	21 (21–21)	0.346	0.026	<0.001	1% / 5%	2.65%	0.827	1.16	-3.631
Student-t copula	17 (17–19)	0.998	0.010	<0.001	0% / 3%	2.77%	0.860	1.10	-3.630
Wavelet t-copula (H1 band)	18 (15–18)	0.807	0.013	<0.001	0% / 3%	2.76%	0.898	1.13	-3.631

Table 6. 1,699 out-of-sample days; 17 breaches expected at 99%. Breach range across Monte Carlo seeds in brackets. Basel shares are the fraction of rolling 250-day windows in the red (≥ 10) or yellow (5–9) zone. Z2 p: Acerbi–Szekely test of ES 97.5% (small = ES too low). Breach loss / ES: average realised loss on VaR-breach days ÷ predicted ES. FZ0: lower is better.

Tails at one day. Same margins, different copula: the Gaussian benchmark breaches its 99% VaR 21 times in every seed, the t-copula 17–19 times, against 17 expected. Its average ES 97.5% is 4.2% below the t-copula's and 10.2% below filtered historical simulation, and it is the only copula model that reaches the Basel red zone (1.4% of windows). All copula models pass Kupiec and the Acerbi–Székely test. On the joint FZ0 score the copula models are statistically tied (t vs Gaussian DM $p = 0.921$), while all beat historical simulation decisively ($p \leq 0.014$): at one day volatility dynamics dominate and dependence is a second-order, tail-only effect. Using the H1 wavelet band instead of daily residuals (WT) changes little (WT vs t: DM $p = 0.753$), which is what Section 4 predicts: daily dependence is the H1 dependence.

What still fails. Breaches cluster: the dynamic-quantile test rejects every model at 99% and Christoffersen's independence test every model except filtered HS ($p = 0.071$), driven by COVID (Table 7). The McNeil–Frey test finds realised breach losses about 10%–16% above predicted ES. A model re-estimated monthly on four years of data cannot anticipate a regime change of the size of March 2020.

Period	Days	HS 99% / 95%	FHS 99% / 95%	G 99% / 95%	T 99% / 95%	WT 99% / 95%
COVID crash (Feb–Jun 2020)	94	8.5 / 18.1	4.3 / 10.6	7.4 / 9.6	6.4 / 9.6	7.4 / 9.6
2022 rate shock	251	3.2 / 8.4	0.4 / 6.4	1.6 / 7.2	1.2 / 7.2	1.2 / 7.6
Other days	1354	0.7 / 2.9	0.5 / 3.3	0.7 / 3.7	0.6 / 3.8	0.6 / 3.8

Table 7. Breach rates (%) by sub-period; targets 1% and 5%.

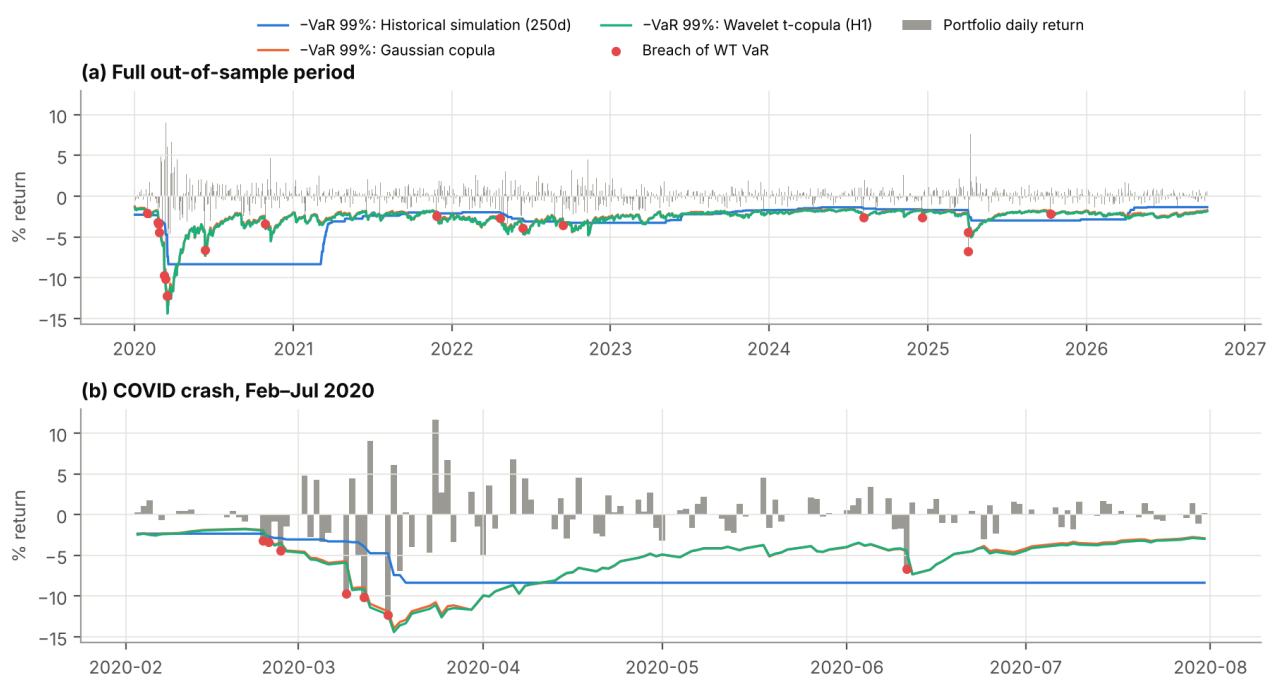


Figure 4. Out-of-sample 99% VaR (negative, % of NAV) and daily portfolio returns. Panel (b) zooms into the COVID crash; red dots are breaches of the wavelet t-copula VaR.

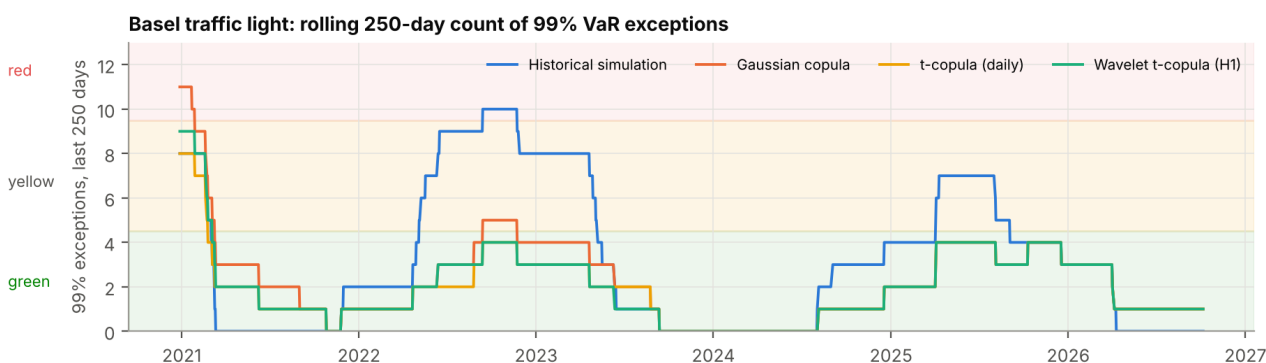


Figure 5. Rolling 250-day count of 99% VaR breaches against the Basel traffic-light zones.

5.2 10-day risk: where the horizon matters

Regulators and asset managers need multi-day risk (FRTB uses a 10-day base horizon). The usual shortcut is $\sqrt{10} \times$ 1-day risk. Table 8 replaces one simplification at a time on 169 non-overlapping 10-day windows, measuring each model's ES 97.5% against the full horizon-aware model, the wavelet empirical copula on the H2 band.

Model	Ignores	Mean ES 97.5%	vs full model: all / stressed	Seed range	99% / 95% breaches	Breach loss / ES	FZ0 loss
Gaussian 1-day $\times \sqrt{10}$	tails; uses $\sqrt{10}$ rule	8.35%	-6.5% / -8.8%	-7.4% to -6.8%	4 / 7	1.19	-2.168
Gaussian copula, simulated paths	tail dependence	7.81%	-12.2% / -14.9%	-13.5% to -12.8%	4 / 9	1.31	-2.077
t-copula, simulated paths	compounds daily draws	7.99%	-10.1% / -12.6%	-11.0% to -10.7%	4 / 9	1.26	-2.105
Daily t-copula imposed at 10 days	asymmetry; band	8.30%	-6.7% / -8.7%	-7.5% to -7.3%	4 / 9	1.22	-2.112
Wavelet t-copula (H2 band)	asymmetry (symmetric t)	8.31%	-6.7% / -8.4%	-7.6% to -7.3%	4 / 9	1.21	-2.121
Daily empirical copula at 10 days	band (uses daily data)	9.07%	+0.9% / +1.9%	+0.9% to +1.3%	4 / 8	1.11	-2.153
Wavelet empirical copula (H2 band)	nothing (full model)	8.96%	reference	reference	4 / 8	1.16	-2.119

Table 8. 169 non-overlapping 10-day windows, 2020–2026 (expected breaches 1.7 at 99%, 8.5 at 95%). 'Stressed' = windows in the top quartile of the full model's ES.

Interpretation. (i) Dropping tail dependence (Gaussian paths) lowers 10-day ES by 12.2% on average and 14.9% when risk is high, and this model has the largest breach losses relative to ES (1.31 $\times$). (ii) Building 10-day risk by compounding a daily t-copula still leaves 10.1%: with independent daily draws the joint tail is diluted, while Section 4 shows the observed crash tail is not. Imposing the daily t-copula directly at the 10-day horizon removes that dilution (-10.1% $\rightarrow$ -6.7%; the H2-band t-copula gives -6.7%). (iii) The remaining gap is the shape of the tail: the empirical copula, which keeps the crash/rally asymmetry, raises 10-day ES by a further 7.8%. (iv) Estimating the copula on the H2 band instead of daily data changes little (daily empirical copula vs full model: +0.9%), which is what Section 4 predicts: crash co-movement at 8–32 days is statistically the same as at the daily scale. The wavelet analysis is what justifies carrying daily crash dependence to the 10-day horizon. (v) The $\sqrt{10}$ rule (-6.5%) does better than Gaussian paths only because two errors partly cancel: it ignores tail dependence, but scaling a fat-tailed 1-day distribution by $\sqrt{10}$ overstates the 10-day tail of the same Gaussian model by 6.4% on average, because 10-day returns have thinner tails than daily ones. With 169 windows the backtest has little power: no model's breach count is rejected and the FZ0 differences are not significant (lowest loss: Gaussian 1-day $\times \sqrt{10}$). The two models that compound daily draws are the only ones whose breach losses are significantly above predicted ES (McNeil–Frey $p = 0.039$ and 0.039 respectively; full model $p = 0.083$).

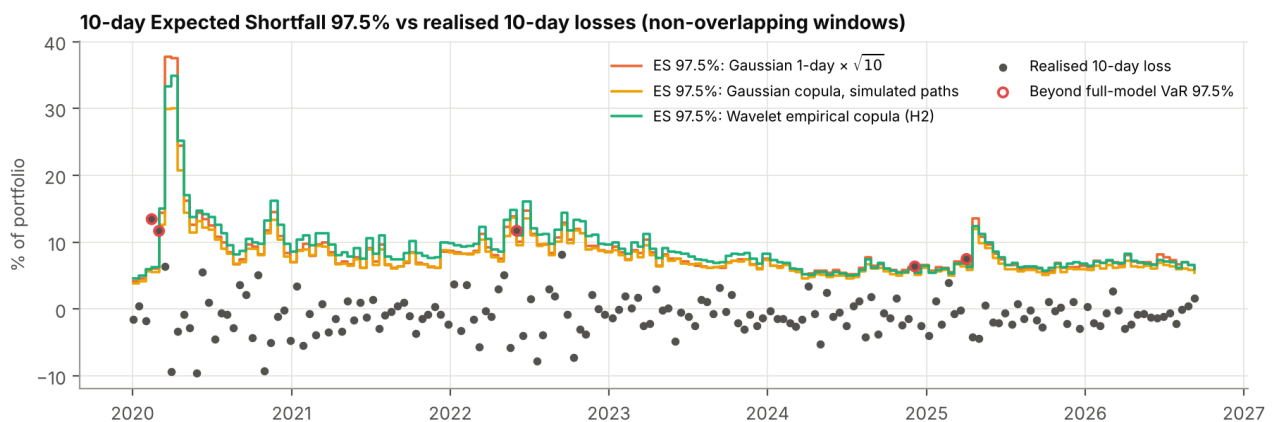


Figure 6. 10-day ES 97.5% forecasts and realised 10-day portfolio losses; circled points exceed the full model's VaR 97.5%.

6. Recommendation for the risk manager

From tomorrow: replace the Gaussian copula and the $\sqrt{10}$ rule with tail-aware, horizon-matched ES.

1. **Daily limit.** Compute 1-day ES 97.5% from the t-copula on daily-updated GARCH margins: 1.94% of NAV today, against 1.78% from the Gaussian copula. Over 2020–2026 the Gaussian model breached its 99% VaR 21 times against 17 expected and was the only copula model to reach the Basel red zone.
2. **10-day limit.** Size the 10-day ES from the wavelet empirical-copula model, not $\sqrt{10} \times 1$ -day: 6.01% of NAV today ($\sqrt{10} \times$ Gaussian: 6.20%; Gaussian paths: 5.38%). Today the shortcut happens to be close, but over 2020–2026 it was below the full model in 76% of windows, by up to 21%, and the Gaussian path model was below it by 15% on average when risk was high.
3. **Diversification.** Do not count on sector diversification in a sell-off at any horizon: crash co-movement beyond a month is statistically the same as at 2–8 days, while rally co-movement halves. Hedge with instruments outside the equity sector set, not with sector rotation.
4. **Trigger.** Escalate to the risk committee when the 250-day count of 99% breaches reaches 5 (Basel yellow). Keep the regulatory capital model unchanged until independent validation; run this as an internal overlay.

7. Robustness and limitations

Robustness. Every out-of-sample number was recomputed with 3 more Monte Carlo seeds (ranges in Tables 6 and 8); conclusions do not change. Results are reported at two tail levels ($q = 5\%$ and 10%); the conclusions hold at both (files in outputs/). Figure 7 shows the rolling t-copula tail dependence: it rises in stress (2020, 2022) at every scale, with the H2 band the most variable.

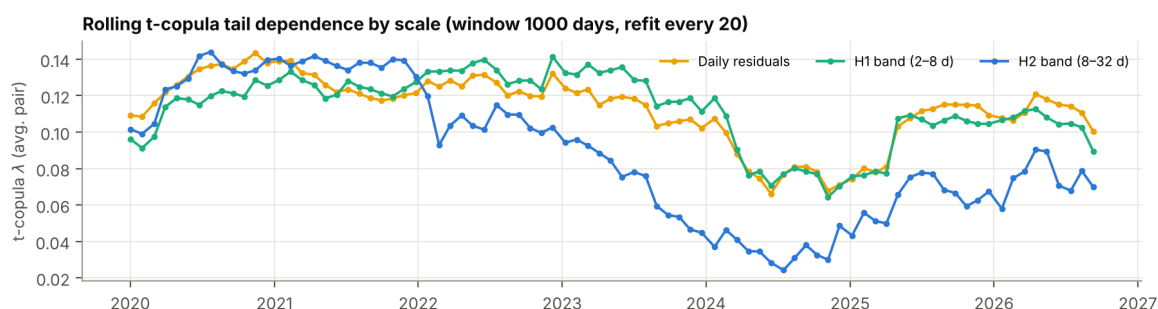


Figure 7. Rolling t-copula tail dependence by scale (window 1000 days, re-estimated every 20 days).

Limitations. (1) Long-scale bands are highly autocorrelated, so H3 has few effective observations; bootstrap intervals widen accordingly but remain approximate. (2) The 10-day backtest has low power (169 windows). (3) The t-copula has one v for all pairs and cannot be asymmetric; the empirical copula fixes asymmetry but cannot extrapolate beyond the worst event in its window. (4) Breach clustering remains: a regime-switching or dynamic (DCC/GAS) copula is the natural next step. (5) Results are for one equal-weight US sector portfolio; the code is portfolio-agnostic.

Further work. Asymmetric parametric copulas per scale (skew-t, vine copulas), dynamic copulas for breach clustering, 10-day ES on overlapping windows with HAC-adjusted tests, and the same study on the Colombo Stock Exchange with corrections for non-synchronous trading.

Appendix A. Reproducibility and use of AI tools

Reproduce. Everything runs locally with no downloads beyond the Python packages. Run `pip install -r requirements.txt` followed by `python -m src.run_all`. This one command regenerates every table, figure and number in this report from the bundled price file (data/etf_prices.csv; src/data.py re-downloads it from Yahoo Finance with `force=True`). All randomness is seeded. The report text reads its numbers from outputs/results.json; none are typed by hand.

AI use. All cognitive and statistical work in this study was done by Team Gmora: framing the research question, choosing the portfolio, data and sample split, selecting the margin, wavelet and copula models, designing the out-of-sample experiments and validation tests, interpreting the results, and forming the recommendation. AI assistants (Anthropic Claude) were used as tools under the team's direction for writing and debugging code, and reviewing code for errors. The team checked every result and can explain every line of code and every claim in this report.

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